whatever, proponents of this orientation hold, for a manned space program. Where in this spectrum of policy options should we be?

Ever since the United States beat the Soviet Union to the Moon, a coherent, widely understood justification for humans in space seems to have vanished. Presidents and Congressional committees puzzle over what to do with the manned space program. What is it for? Why do we need it? But the exploits of the astronauts and the moon landings had elicited—and for good reason—the admiration of the world. It would be a rejection of that stunning American achievement, the political leaders tell themselves, to back off from manned spaceflight. Which President, which Congress wishes to be responsible for the end of the American space program? And in the former Soviet Union a similar argument is heard: Shall we abandon, they ask themselves, the one remaining high technology in which we are still world leaders? Shall we be faithless heirs of Konstantin Tsiolkovsky, Sergei Korolev, and Yuri Gagarin?

The first law of bureaucracy is to guarantee its own continuance. Left to its own devices, without clear instructions from above, NASA gradually devolved into a program that would maintain profits, jobs, and perquisites. Pork-barrel politics, with Congress playing a leading role, became an increasingly powerful force in the design and execution of missions and long-term goals. The bureaucracy ossified. NASA lost its way.

On July 20, 1989, the twentieth anniversary of the *Apollo 11* landing on the Moon, President George Bush announced a long-term direction for the U.S. space program. Called the Space Exploration Initiative (SEI), it proposed a sequence of goals including a U.S. space station, a return of humans to the Moon, and the first landing of humans on Mars. In a later statement, Mr. Bush set 2019 as the target date for the first footfall on that planet.

And yet the Space Exploration Initiative, despite clear direction from the top, foundered. Four years after it was mandated, it did not even have a NASA office dedicated to it. Small and inexpensive lunar robotic missions—that otherwise might well have been approved—were canceled by Congress because of guilt by association with SEI. What went wrong?